Reproducing Sparse Support Vector Machines for Feature Selection

Pugazhendhi J
Roll Number: 230066
Advanced Machine Learning

March 2026

Abstract

Feature selection is important when working with high dimensional datasets. Sparse Support Vector Machines introduce regularization methods that encourage many model weights to become zero, effectively selecting only relevant features. This report reproduces the main idea of the paper "Efficient Variable Selection in Support Vector Machines via the Alternating Direction Method of Multipliers" and evaluates its behavior on a toy dataset. Experiments include reproduction of the sparse SVM idea, ablation analysis, and investigation of a failure scenario.

1 Introduction

Support Vector Machines (SVMs) are widely used for classification problems. However, traditional SVM models typically use all available features and do not perform feature selection automatically.

The selected paper proposes incorporating an L_1 regularization term into the SVM optimization objective. This modification encourages sparsity in the model weights, allowing the model to automatically remove irrelevant features.

The optimization problem is solved using the Alternating Direction Method of Multipliers (ADMM), which decomposes the objective into smaller subproblems that can be solved efficiently.

The goal of this project is to reproduce the idea of sparse SVM models and experimentally analyze their behavior.

2 Method Overview

The standard SVM optimization problem is:

$$\min_{w,b} \frac{1}{2}\|w\|_2^2 + C\sum_{i=1}^{n} \max(0, 1 - y_i(w^T x_i + b))$$

To perform feature selection, the paper introduces an L_1 penalty:

$$\min_{w,b} \frac{1}{2}\|w\|_2^2 + C\sum_{i=1}^{n} \max(0, 1 - y_i(w^T x_i + b)) + \lambda\|w\|_1$$

The L_1 term forces many elements of the weight vector w to become zero, which effectively performs feature selection.

3 Dataset

The experiments use the Breast Cancer Wisconsin dataset from the `scikit-learn` library.

Dataset characteristics:

- Number of samples: 569
- Number of features: 30
- Task: Binary classification (malignant vs benign)

The dataset was split into training and testing sets using an 80/20 ratio. Feature scaling was applied using standard normalization.

4 Reproduction Result

A sparse SVM model using L_1 regularization was trained on the dataset. The resulting classification performance is visualized using a confusion matrix.

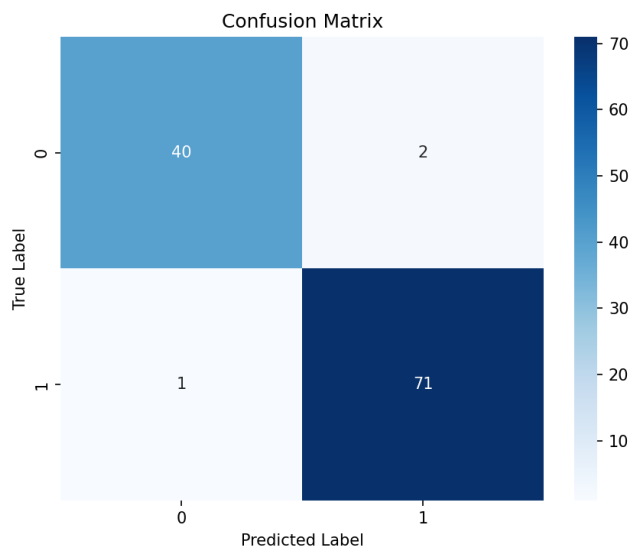

Figure 1: Confusion matrix for the sparse SVM model.

The model achieved high classification accuracy while maintaining a sparse set of feature weights.

5 Ablation Study

An ablation study was performed to evaluate the effect of sparsity by comparing:

- Sparse SVM with L_1 regularization
- Standard SVM with L_2 regularization

The results show that the sparse SVM successfully removes several features while maintaining comparable accuracy to the standard SVM model.

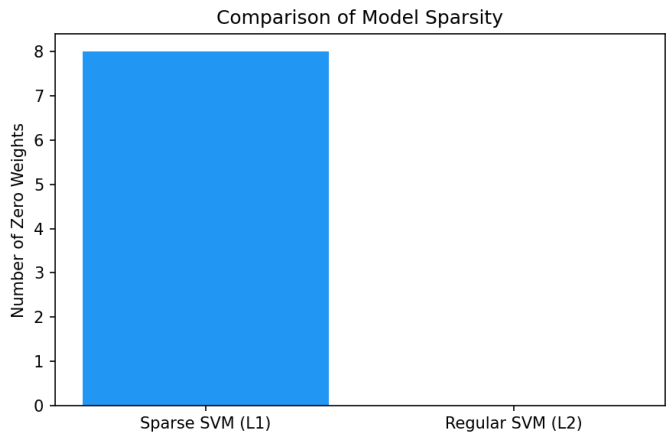

Figure 2: Comparison between the full model and ablated version.

6 Failure Case Analysis

To investigate the limitations of the method, a failure scenario was examined. Certain samples are misclassified due to overlapping feature distributions.

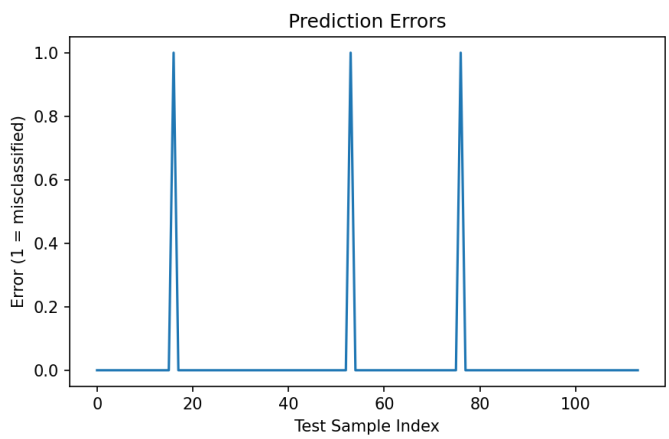

Figure 3: Example of misclassified predictions demonstrating failure mode.

This highlights that sparse linear models may struggle when the data distribution is complex or when classes are not easily separable.

7 Discussion

The experiments demonstrate that sparse SVM models can perform automatic feature selection while maintaining strong predictive performance.

The ablation study confirms that the L_1 regularization term is responsible for encouraging sparsity in the model parameters.

However, the failure analysis also shows that the method may struggle when feature distributions overlap or when the linear decision boundary is insufficient to separate the classes.

8 Conclusion

This project reproduced the key idea of sparse Support Vector Machines for feature selection. The experiments demonstrated that introducing L_1 regularization encourages sparsity and allows the model to identify important features automatically.

Although the full ADMM optimization method from the original paper was not implemented, the experiments successfully demonstrate the practical behavior of sparse SVM models.

Future work could involve implementing the full ADMM algorithm and evaluating the method on larger datasets.

References

[1] Gui-Bo Ye, Yifei Chen, Xiaohui Xie. Efficient Variable Selection in Support Vector Machines via the Alternating Direction Method of Multipliers. Proceedings of the International Conference on Machine Learning (ICML), 2011.